

Electricity pylons

These carry power lines that transmit electricity from the power station to electricity users.

Generator

The generator converts energy from the turbines into electricity.

Cooling towers

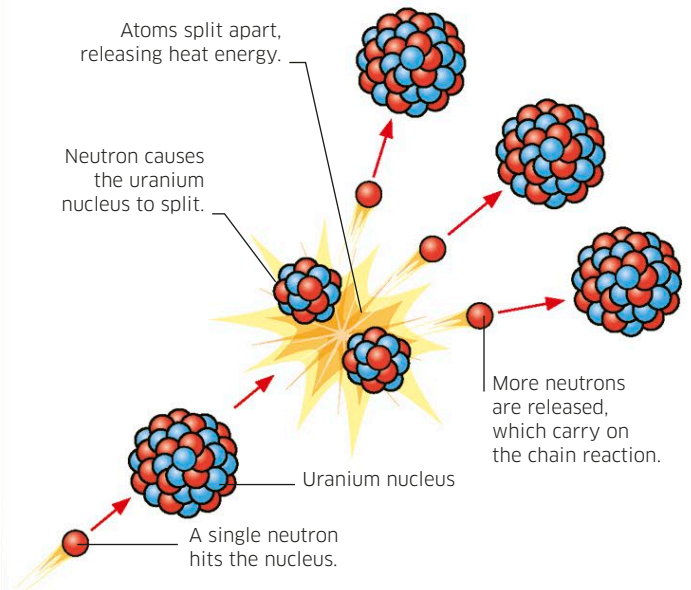
Large towers receive the steam and condense it back into water.

Condenser loop

Cooled water is pumped back to the turbine, ready to be heated again.

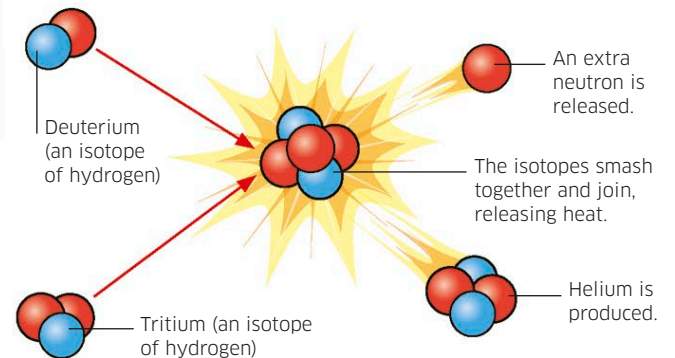
Nuclear fission

The nuclei of atoms can split apart or join together, forming new elements and releasing energy. A large atomic nucleus splitting in two is called nuclear fission. A neutron hits the nucleus of a uranium atom, causing it to split, or fission, in two. More neutrons are released as a result, and these hit more nuclei, creating a chain reaction. The extra energy that is released ends up as heat that can be used to generate electricity.



Nuclear fusion

The process in which two smaller atomic nuclei join together is called fusion. Two isotopes of hydrogen are smashed into each other to make helium, releasing heat energy and a spare neutron. Fusion takes place in stars, but has not yet been mastered as a viable form of producing energy on Earth, due to the immense heat and pressure needed to start the process.



Alpha radiation

Some large nuclei release a positively charged particle made of two protons and two neutrons, called an alpha particle.



Beta radiation

In some nuclei, a neutron changes to a proton, creating an electron called a beta particle, which shoots out of the nucleus.

Gamma radiation

Gamma rays are electromagnetic waves released during alpha and beta decay.

Containing radiation

Radiation can be extremely harmful to human health and containing it can be tricky. Alpha, beta, and gamma radiation can pass through different amounts of matter because they have different speeds and energy. Alpha particles can be stopped by just a sheet of paper, or skin. Beta rays can pass through skin but not metal. Gamma rays can only be stopped by a sheet of lead or thick concrete.

